

Lab 5: Ensemble Learning

Evaluation: 5% of the total grade

Objective:

This lab exercise focuses on building a Stacking Classifier using Logistic Regression as the meta-model, with the base models extended to include Random Forest, Gradient Boost, Extra Trees, and AdaBoost. Students will compare the performance of individual models against the ensemble model and answer conceptual questions related to ensemble learning as follows:

1. What is the main advantage of using ensemble learning methods like stacking compared to using a single classifier?
2. Why do we use a Logistic Regression model as the meta-learner in stacking? What role does it play?
3. How do base models with different strengths and weaknesses contribute to the performance of the stacking model?
4. When should we use hard voting versus soft voting in ensemble learning?
5. How does adding more diverse base models, such as AdaBoost and Random Forest, affect the performance of the stacking classifier?

Step-by-step Instructions:

Step 1: Load and explore the Titanic (version 1) dataset from openml.org used in Lab 1

Step 2: Define the Base Models and Meta-Model for Stacking

In this step, we introduce base learners such as Gradient Boost, Extra Trees, Random Forest, and AdaBoost, as well as the meta-model such as Logistic Regression.

Step 3: Train and evaluate the Stacking Classifier

Step 4: Compare with Individual Models

Train and evaluate each base model individually to compare their performance with the stacking ensemble. Evaluate the performance of the stacking classifier and answer the following questions:

1. How does it compare to the performance of the individual models?
2. Which individual model performed the best? Explain why this model might be performing well on this dataset.
3. Compare the performance of the stacking classifier with that of the best individual model. Is there an improvement? If not, why do you think stacking did not perform better?

Submit your lab report including your completed Jupyter notebook with the results of all tasks and answers to the questions by June 24, 2025.

Lab Grading Criteria

Fraction	0.8 – 1	0.4 – 0.7	0 – 0.3	Max Score
Category				
Model development workflow	Followed all steps in the modeling process	Followed most steps in the modeling process	No order in the modeling process	1
Tasks and observations	Performed all the tasks correctly, results properly presented and well explained	Performed most of the tasks correctly, results fairly presented and explained	Performed some of the tasks correctly, poorly presented results with little explanation	3
Code	Well organized, easy to follow and clean code with meaningful comments	Somewhat organized, a little hard to follow and fairly clean code with some comments	Not organized, very hard to follow and messy code with little or no comments	1
Total				5